**Vision of the Department**

*To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.*

**Mission of the Department**

*To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

**Session 2025-2026**

| | |
|---|---|
| **Vision:** Dream of where you want. | **Mission:** Means to achieve Vision |

**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

| | | | |
|---|---|---|---|
| PEO1 | **Preparation** | **P: Preparation** | **Pep-CL abbreviation pronounce as Pep-si-lL easy to recall** |
| PEO2 | **Core Competence** | **E: Environment (Learning Environment)** | |
| PEO3 | **Breadth** | **P: Professionalism** | |
| PEO4 | **Professionalism** | **C: Core Competence** | |
| PEO5 | **Learning Environment** | **L: Breadth (Learning in diverse areas)** | |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

**Keywords of POs:**
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

**Name and Signature of Student and Date**
(Signature and Date in Handwritten)

**Vision of the Department**

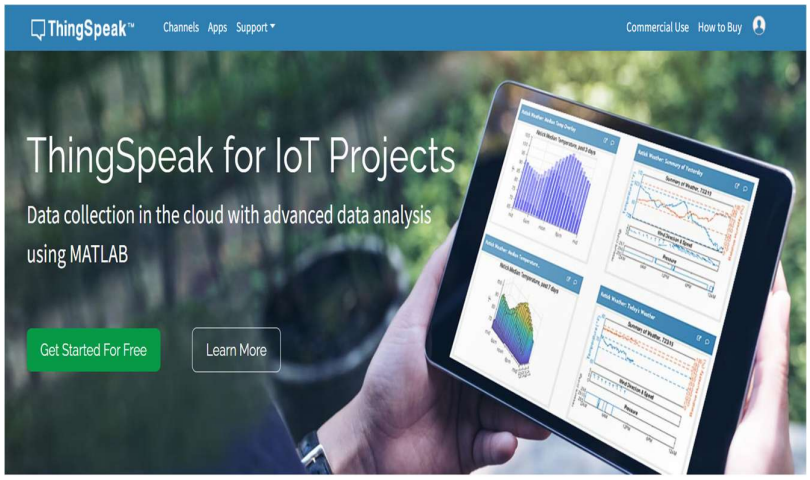
*To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.*

**Mission of the Department**

*To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

| **Session** | **2025-26 (ODD)** | **Course Name** | **PE-I - Geo-Intelligence for Smart IoT Devices Lab** |
|---|---|---|---|
| **Semester** | **5** | **Course Code** | **23IOT1523** |
| **Roll No** | **64** | **Name of Student** | **Sheikh Wasimuddin** |

| | |
|---|---|
| Practical Number | **10** |
| Course Outcome | Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets. |
| Aim | Show visualization of any IOT related dataset on ThingSpeak.<br>Create appropriate channel and fields |
| Problem Definition | To create a ThingSpeak channel that stores smart home IoT sensor data such as temperature, humidity, pressure, and visualize it through charts for better understanding and monitoring |
| Theory<br>(100 words) | ThingSpeak is an IoT analytics platform that allows users to collect, analyze, and visualize sensor data in real time. It provides cloud-based storage and built-in visualization tools to help monitor and interpret data from IoT devices. In this project, a ThingSpeak channel is created to store smart home sensor data such as temperature, humidity, and pressure, which are commonly monitored environmental parameters. Each of these parameters is assigned to a specific field within the channel — for instance, Field 1 for temperature, Field 2 for humidity, and Field 3 for pressure.<br>Data from IoT sensors (like DHT11, BMP180, or BME280) is periodically sent to the ThingSpeak channel using an API key through platforms like Arduino, NodeMCU, or Raspberry Pi. Once uploaded, ThingSpeak automatically generates real-time visualizations such as line charts for each field, allowing users to observe patterns and trends over time. These visual insights help in smart home applications — for example, adjusting ventilation based on humidity or controlling heating systems using temperature readings. Thus, ThingSpeak provides an easy and effective way to collect, monitor, and analyze IoT sensor data for smart home automation and environmental monitoring. |

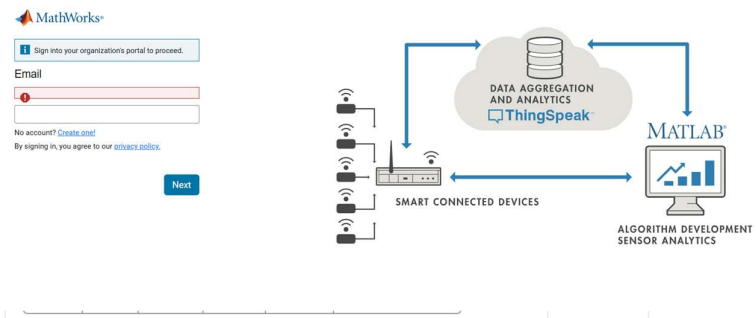
| | |
|---|---|
| Procedure and Execution<br><br>(100 Words) | Implementation Steps:<br>1. Open the ThingSpeak website.<br>2. Log in using your MathWorks account.<br>3. Click "New Channel" to create a new IoT data channel.<br>4. Add fields for Temperature (Field 1), Humidity (Field 3), Pressure (Field 8).<br>5. Save the channel after enabling the required fields.<br>6. Go to Data Import/Export → Import Data.<br>7. Upload the prepared CSV file thingspeak_smart home iot_data.csv.<br>8. Verify that the created_at column and field values are mapped correctly.<br>9. Click Import to load the data into the channel.<br>10. Open the Private View tab to see auto-generated charts for each field.<br>11. Customize each chart by setting proper titles, labels, and colors.<br>12. Analyze the visualized smart home IoT data for variations and patterns over time.<br><br>  <br><br> |

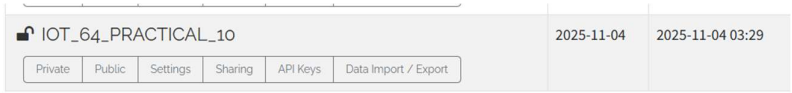
**Vision of the Department**

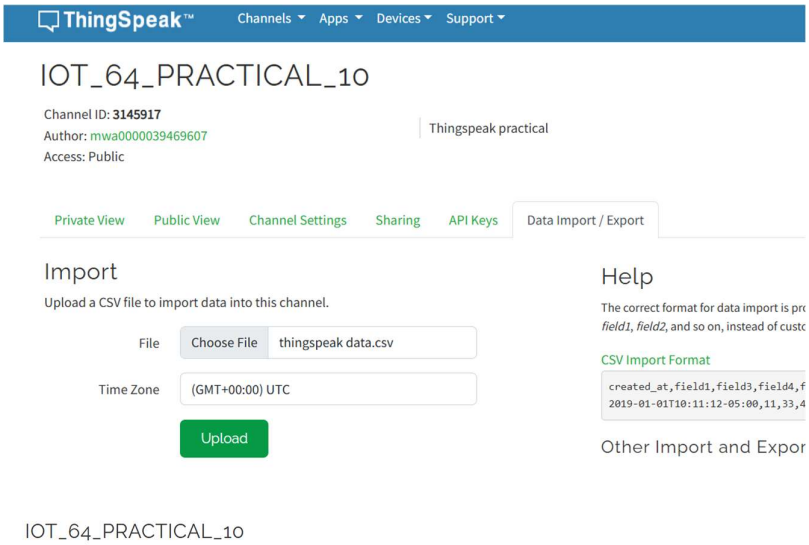
*To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.*

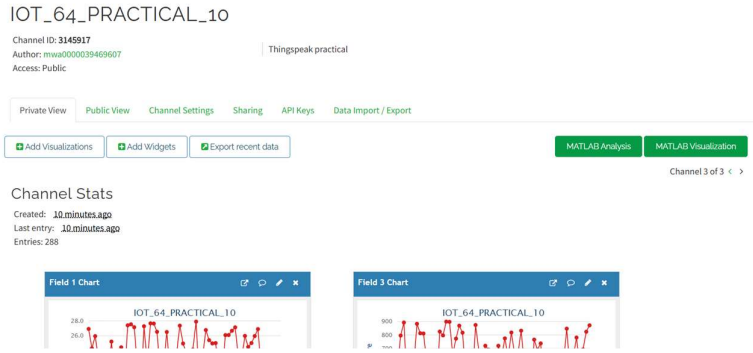
**Mission of the Department**

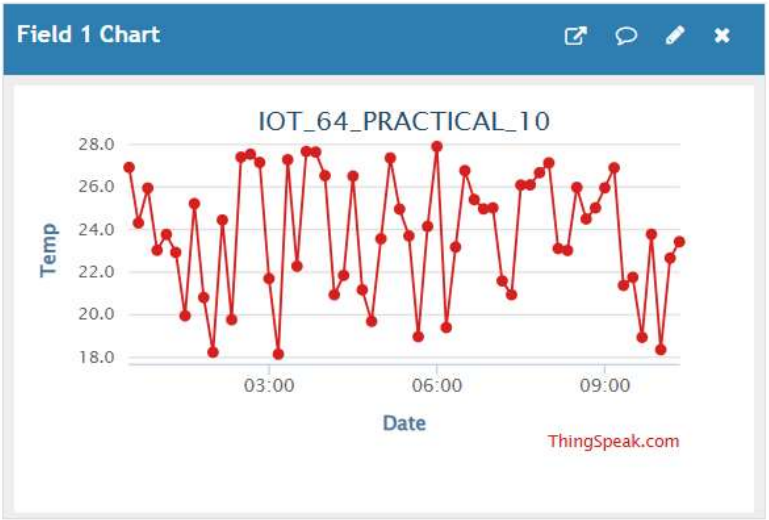
*To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

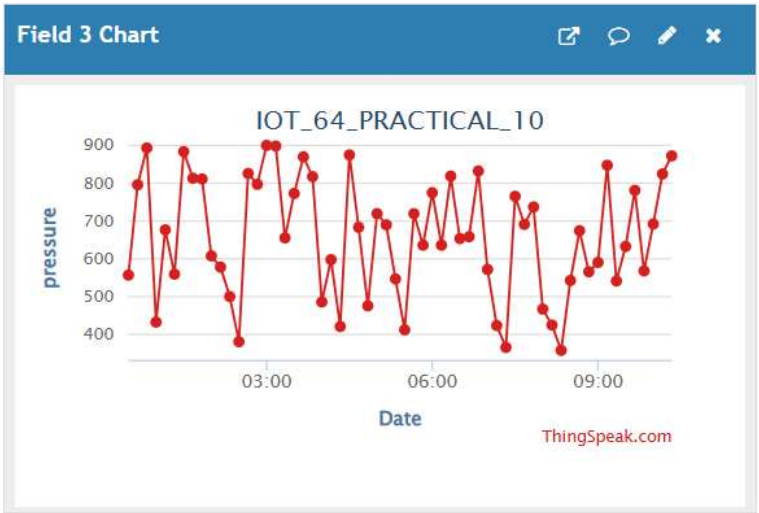


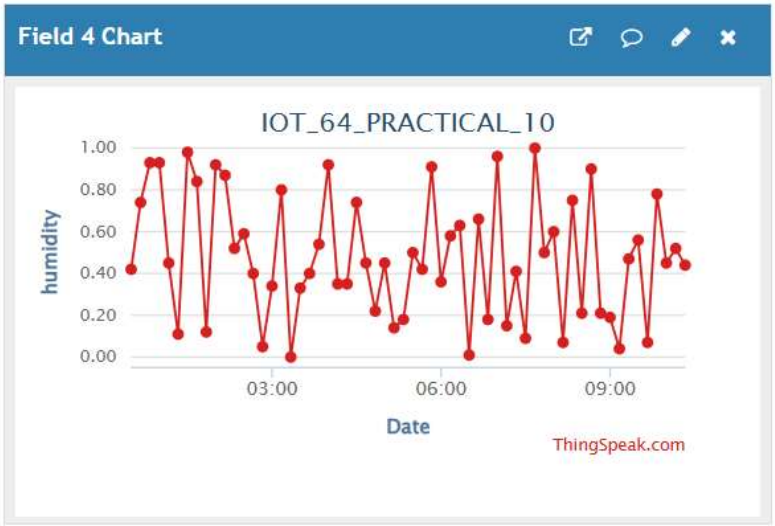




Field 1 Chart
IOT_64_PRACTICAL_10
Temp
28.0
26.0
24.0
22.0
20.0
18.0
03:00
06:00
09:00
Date
ThingSpeak.com


Field 3 Chart
IOT_64_PRACTICAL_10
pressure
900
800
700
600
500
400
03:00
06:00
09:00
Date
ThingSpeak.com

| | |
|---|---|
| |  |
| Output Analysis | After successfully creating the ThingSpeak channel and configuring the fields for Temperature, Humidity, and Pressure, real-time data from IoT sensors is uploaded to the platform. The output appears as dynamic charts that continuously update as new sensor readings are received.The Temperature graph shows variations in room temperature over time, helping to identify patterns such as daily heating or cooling trends. The Humidity chart displays changes in moisture levels, which can indicate ventilation efficiency or comfort conditions within the home. The Pressure graph reveals atmospheric pressure changes, which can be useful for weather prediction or air quality monitoring.Overall, the ThingSpeak dashboard visually represents all three parameters in separate line plots, making it easy to monitor environmental conditions at a glance. This real-time visualization enables better decision-making, early detection of anomalies (like sudden temperature spikes), and efficient management of smart home systems through data-driven insights. |
| Conclusion | The implementation of an IoT-based smart home data visualization system using ThingSpeak successfully demonstrates how sensor data can be collected, stored, and analyzed in real time. By creating a dedicated channel with fields for temperature, humidity, and pressure, the project provides an effective way to monitor environmental parameters remotely. The graphical representation of data on ThingSpeak enhances understanding of trends and variations, enabling timely responses to changes in home conditions. Overall, this project highlights the importance of IoT |

| | |
|---|---|
| | platforms like ThingSpeak in building intelligent, data-driven smart home systems that improve comfort, efficiency, and environmental awareness. |
| Plag Report (Similarity index < 12%) | |
| Date | **28/10/2025** |